IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

POWER ELECTRONICS AND POWER SYSTEMS

Friday, 9 May 2025, 2:30 PM

Time allowed: 2:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equation for buck SMPS in continuous mode $\frac{V_O}{V_I} = \delta$

Voltage ratio equations for boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{1}{1-\delta}$

Voltage ratio equations for buck-boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{-\delta}{1-\delta}$

Three-Phase Systems (Balanced)

Line/Phase Voltages and Currents

Star $V_L = \sqrt{3}\, V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\, I_P$

Power $P_{3\phi} = 3\, V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Transformers

Ideal transformer $\bar{V}_2 = k_N \bar{V}_1$; $\bar{I}_2{}' = k_N \bar{I}_2$

Impedances referral $\bar{Z}'_2 = {\bar{Z}_2}/{k_N{}^2}$

Voltage regulation $\Delta V_2 = \frac{\left|V_2^{o/c}\right| - |V_2^{load}|}{\left|V_2^{o/c}\right|}$

Induction Machines

$\omega_S = {\omega_E}/{P}$; $\quad s = \frac{\omega_S - \omega_R}{\omega_S}$; $\quad T_{EM} = \frac{3\, I'^2_R\, R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi\, r_C^2} + \frac{1}{\sigma_C 2\pi\, r_O\, t_O}$ $\quad L' = \frac{\mu}{2\pi} ln\left({r_O}/{r_C}\right)$ $\quad C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({r_O}/{r_C}\right)}$ $\quad G' = \frac{2\pi\sigma_I}{ln\left({r_O}/{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2}$ $\quad L' = \frac{\mu}{\pi} ln\left({d}/{r_C}\right)$ $\quad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({d}/{r_C}\right)}$ $\quad G' = \frac{\pi\sigma_I}{ln\left({d}/{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}}\cos(\theta) - \frac{V_R V_S}{Z_{SR}}\cos(\theta+\delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}}\sin(\theta) - \frac{V_R V_S}{Z_{SR}}\sin(\theta+\delta)$$

Power Flow assuming inductance-dominated impedance

$$P_S \simeq \frac{V_R V_S}{X_{SR}}\sin(\delta) \qquad Q_S \simeq \frac{V_S}{X_{SR}}\left(V_S - V_R\cos(\delta)\right)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\, P_S + X\, Q_S}{|V_S|}$

1. This question addresses transformers [25]

a) Some written questions on transformers

i) What is it that a transformer does that is highly beneficial in a power system? [1]

ii) Why is it beneficial? [1]

iii) How does it do it? [1]

iv) Most transformers in the world today are step down – why is this? [1]

b) A 50 Hz 100 kVA 2.4 kV to 240 V single phase step-down transformer has the following parameters for its approximate equivalent circuit

Winding resistance	$R_{\mathrm{W}} = 0.8\ \Omega$
Winding reactance	$X_{\mathrm{W}} = 5.0\ \Omega$
Magnetizing resistance	$R_M = 7\ \mathrm{k}\ \Omega$
Magnetising reactance	$X_M = 1\ \mathrm{k}\ \Omega$

i) Calculate the turns ratio [1]

ii) Calculate the rated current on the primary side [1]

iii) Calculate the rated current on the secondary side [1]

c) For the transformer in (b), rated voltage is applied at the primary side and the secondary side is open circuit. Answer the questions below.

i) Calculate the voltage on the secondary side [1]

ii) Calculate the winding losses [1]

iii) Calculate the magnetizing losses [1]

iv) Calculate the power factor stating whether it is lagging or leading [2]

v) Calculate the efficiency [1]

d) For the transformer in (b), rated voltage is applied at the primary side and a load of 0.5 + j 0.1 Ω is connected to the secondary side. Answer the questions below.

i) Calculate the voltage on the secondary side [3]

ii) Calculate the current on the secondary side [3]

iii) Calculate the real and reactive power of the load [3]

iv) Calculate the efficiency [3]

2. This question addresses 3 phase, synchronous generators and inverters [20]

a) A 3-phase 500 kVA synchronous generator is star connected to an infinite bus at 6 kV.

The speed of rotation is 1200 revolutions per minute.

i) What is the frequency at the infinite bus? [1]

ii) How many pole pairs does the generator have? [1]

iii) If the generator is generating 450 kW, calculate the Torque [2]

iv) If the torque increases by 5 %, calculate the mechanical speed of the machine and the power being generated. [2]

v) For this increased torque the power factor is unity. Calculate the current [2]

vi) If the load angle δ is 20° and generator emf is 3.5 kV ignoring armature resistance calculate the synchronous reactance and draw the phasor diagram for this generator. [4]

b) A 3-phase permanent magnet synchronous motor (PMSM) is driven by a three-phase inverter.

i) At 3000 RPM, the modulator frequency of the inverter is 50 Hz. What is the modulator frequency if the speed of the PMSM is changed to 600 RPM? [4]

ii) The inverter has dc input voltage (rail-to-rail) of 1000 V, which is equivalent to $V_{DC+}=V_{DC-}=500V$. Write a sinusoidal PWM modulator function to generate a 30Hz inverter voltage with an amplitude of 300 V per phase (you only need to write the modulator for one of the phases). [2]

iii) At 1000RPM, the EMF of the motor is 100V (line-to-line RMS value). The dc input voltage (rail-to-rail) of the inverter is 1000V. What is the modulation ratio of the inverter at a speed of 3000 RPM? [2]

3. This question addresses power flow in lines and cables. [25]

a) Consider the characteristics of single-phase overhead lines.

i) An overhead line is found to have an inductance per unit length of 100 μH/km and is operated at 50 Hz and 290 kV. For a line with a length of 150 km, calculate the maximum power transferred along the line. [5]

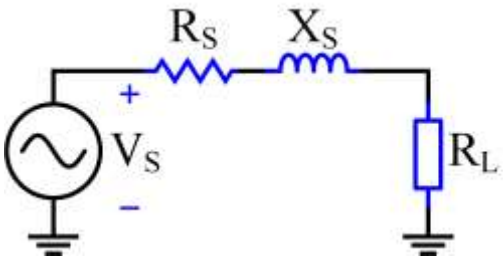
ii) For the line in i), if the maximum load angle is limited to 15°, calculate the maximum current. [5]

iii) An overhead line has a series impedance of 0.65 + j 0.35 Ω. The voltage at the sending end is 15 kV. A real power of +11.0 MW is measured flowing into the line at the sending end. The voltage at the receiving end is measured at 14.3 kV. Estimate the reactive power as seen at the sending end. [5]

b) Consider the characteristics of single-phase overhead lines suppling an impedance load, as shown in the figure below, and answer these questions.

i) The overhead line has an impedance of 0.5 + j4.0 Ω. The voltage at the sending end is 1300 V. If a pure resistive load with impedance of 15.0 Ω is connected at the receiving end, calculate the terminal voltage magnitude at the receiving end, as well as the active power that is consumed by the load. [5]

ii) For the line in i), consider a pure resistive load R_L connected at the receiving end. Calculate the maximum power that can be supplied to the load. (Hint: find the power delivered to R_L as a function of R_L and identify the maximum value of the function.) [5]



4. This question addresses switch-mode power supplies. [30]

a) A Buck SMPS is to be designed to convert 80V to 25V with a maximum output power of 150W. It will operate in continuous mode. The following design choices have been made.

Switching frequency, $f = 20$ kHz
Inductor, $L = 800$ μH
Capacitor, $C = 1000$ μF with an effective series resistance, $R_{ESR} = 40\text{m}\Omega$

i) Calculate the duty cycle needed for the 80V to 25V voltage conversion ratio, neglecting the voltage drop on the MOSFET and diode. [4]

ii) Using the same input voltage and duty cycle as in a), find the minimum output current below which the Buck SMPS would enter discontinuous conduction mode. [4]

iii) Using the same input voltage and duty cycle as in a), and assume the circuit is operating at the maximum power point. Calculate the maximum instantaneous current that flows through the diode. [4]

iv) The MOSFET to be used will be chosen from a range that has the following properties when it is switched:

The turn-on energy loss of the MOSFET is 20.0 μJ
The turn-off energy loss of the MOSFET is 25.0 μJ

It is required that the junction temperature is 70°C or less in operation. The temperature of the ambient air is 40°C and the heatsink is such that the junction-to-air thermal resistance is R_{th}=15K/W.
Calculate the maximum value of the on-state resistance, R_{DS}, that the MOSFET can be allowed to have to meet this thermal design at the maximum output power of 150W. [8]

b) A second SMPS of Buck-Boost form is required to produce a -25 V output from the 80 V input. The circuit will be operated at a switching frequency of 30 kHz in discontinuous conduction mode. For operation at 150 W output power a duty cycle of 0.15 will be used.

i) Calculate the stored energy that should be present in the inductor at the end of the on-time to transfer 150 W under the stated conditions. [2]

ii) For the given input and output voltage settings, calculate the maximum on-time that can be used while remaining in the discontinuous mode. [3]

iii) For operation at 150 W output power with a duty cycle of 0.15, calculate the inductor current at the end of the on-time. [3]

iv) Calculate the output voltage ripple across the capacitor if the effective series resistance of the capacitor is R_{ESR} = 15 mΩ (you may ignore the ripple across the capacitance of the capacitor). [2]